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Effects of aerobic and cognitively-engaging physical activity on academic skills: A cluster randomized controlled trial

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ABSTRACT

This cluster randomized controlled trial (trial-number #) compares effects of two school-based physical activity interventions (aerobic vs. cognitively-engaging) on reading, mathematics, and spelling achievement; and whether effects are influenced by volume of moderate-to-vigorous physical activity and baseline achievement. Twenty-two primary schools participated, where a third and fourth grade class were randomly assigned to the intervention or control group. Intervention groups were randomly assigned to a 14-week aerobic or cognitively-engaging intervention, receiving four physical education lessons a week. Control groups followed their regular physical education program. Academic achievement of 891 children (mean age 9.17 years, 49.4% boys) was assessed with standardized tests before and after the interventions. Post-Test academic achievement did not significantly differ between intervention groups and control group. A higher volume of moderate-to-vigorous physical activity resulted in better post-test mathematics achievement in both intervention groups, and post-test spelling achievement in the cognitively engaging intervention group. Compared to the control group, lower achievers in reading performed better in reading after the cognitively-engaging intervention. A combination of moderate-to-vigorous physical activity and cognitively-engaging exercises seems to have the most beneficial effects. Future intervention studies should take into account quantitative and qualitative aspects of physical activity, and children's baseline academic achievement.

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KEYWORDS
Physical education; exercise; academic performance; primary schools; child development

Introduction

The positive effects of physical activity on children's health, physical fitness, and motor development are well known (Morgan et al., 2013; Wu et al., 2017). It is therefore unfortunate that children often get only few opportunities to be physically active during the school day, mainly because many educators believe that spending time on physical activity interferes with academic achievement (Howie & Pate, 2012). Contrary to what is often believed, there is little evidence for the adverse effects of physical activity on academic performance (Singh et al., 2018), with some studies rather suggesting that physical activity can have beneficial effects on academic performance (De Greeff et al., 2018).

Results on physical activity programs implemented in the school-setting are inconsistent, however, with some studies reporting positive effects of physical activity on academic achievement (Ericsson & Karlsson, 2014), whereas others find mixed (Resaland et al., 2016), or null effects (Ahamed et al., 2007; Coe et al., 2006). These inconsistent findings might be due to the wide variation in the type and intensity of the physical activities implemented in different studies (Donnelly et al., 2016). Most studies examining effects of school-based physical activity programs have focused on quantitative aspects (i.e., duration, frequency, and intensity), providing evidence for the effectiveness of aerobic physical activity at a moderate to vigorous intensity level (MVPA; Coe et al., 2006; Mura et al., 2015), and indications of dose–response effects (Davis et al., 2011; Mura et al., 2015). Studies mostly focus on cognitive outcomes, however, finding the strongest effects of MVPA for executive functioning (EF) (Álvarez-Bueno et al., 2017), the cognitive functions that guide and control goal-directed behaviour (Diamond, 2012). The positive effects on EF are thought to be brought about via changes in the brain, such as an upregulation of growth factors and monoamines, increased cerebral blood flow, neurogenesis, and improved brain functioning, mainly in brain areas also involved in EF (Best, 2010; Lubans et al., 2016). As EFs are closely related to academic achievement (Diamond, 2012), it has been suggested that physical activity's effects on academic achievement are a result of improved EFs (Donnelly et al., 2016). Therefore, characteristics of physical activity that are beneficial for EF, such as intensity and dose, can be expected to also aid academic performance, although more research is needed to substantiate this claim.

Other researchers have focused on qualitative aspects (i.e., type) of physical activity (Crova et al., 2014; Pesce, 2012). Although this line of research has not yet examined effects of different types of physical activity on academic achievement, some first evidence indicates that EFs benefit more from

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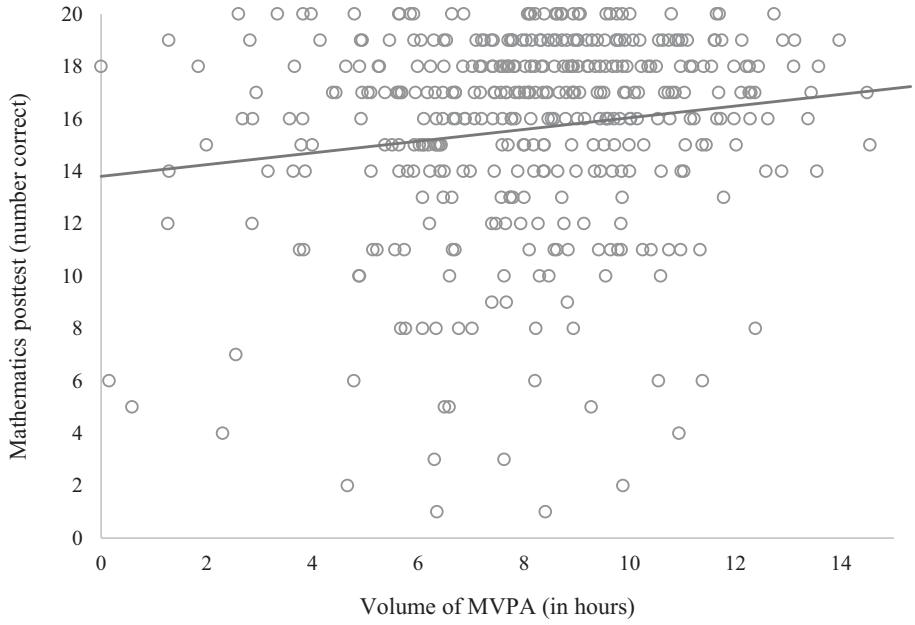

Figure 1. The relation between volume of MVPA and mathematics post-test scores for the two intervention groups.

cognitively engaging physical activity, compared to "simple" repetitive activities involved in aerobic physical activity (Crova et al., 2014; Pesce, 2012). Cognitive engagement is defined as "the degree to which the allocation of attentional resources and cognitive effort is needed to master difficult skills" (Tomporowski et al., 2015). This type of physical activity is for example, seen in team sports, where children have to focus their attention, plan a strategy, collaborate with team mates, and so on. It is argued that cognitively engaging physical activity requiring complex, controlled and adaptive cognition, activates brain areas that are also involved in higher-order cognitive functioning, thereby aiding cognitive development (Álvarez-Bueno et al., 2017; Lubans et al., 2016). Considering the close link between EF and academic achievement (Diamond & Lee, 2011; Howie & Pate, 2012; Lubans et al., 2016), it can be hypothesized that this type of physical activity will be beneficial for academic achievement as well. Research focusing on academic outcomes is needed to confirm this hypothesis.

The mixed results on the effectiveness of physical activity interventions can also be attributed to the fact that not all children benefit to the same extent, since several studies suggest that physical activity has the most beneficial effects on cognition (Diamond & Lee, 2011; Drollette et al., 2014; Sibley & Beilock, 2007) and academic achievement (Resaland et al., 2016) for children with the poorest performance at baseline, possibly because they have most room for improvement. These children are a vulnerable population, being at risk of school drop-out due to their cognitive difficulties (Rumberger & Lim, 2008), making it of great importance to examine whether physical activity can provide an effective mean for improving their academic performance.

The present study

It remains yet unknown whether the effectiveness of physical activity interventions on academic achievement depends on the type of physical activity involved. Therefore, the present study examines the causal effects of two school-based physical activity interventions on primary school children's achievement in reading, mathematics, and spelling. One intervention focuses on aerobic physical activity at an MVPA level, the other intervention on cognitively engaging physical activity. Second, to get a better understanding of characteristics that influence intervention effectiveness, it is examined whether the volume of MVPA is related to intervention effectiveness. As previous studies have provided evidence for MVPA (Coe et al., 2006; Mura et al., 2015) and dose–response effects (Davis et al., 2011; Mura et al., 2015), larger effects are expected for children who are exposed to a higher volume of MVPA. Third, it is examined whether children's prior level of achievement is related to the intervention effects, with the expectation of larger effects for lower-achieving children at baseline.

Materials and methods

Design and participants

This study is part of the "Learning by Moving" project, a cluster randomized controlled trial (RCT) examining the effects of two physical activity interventions. A cluster power analysis with.40 as effect size (Davis et al., 2011; Sibley & Etnier, 2003) resulted in a required sample of ≥40 classes (20 schools), with 25 children per class (power 0.90, intraclass correlation ρ = .10, 1-tailed, a = .05). Randomization was performed by an independent researcher using numbered containers.

Twenty-four regular primary schools were recruited in the school year 2015/2016 and were matched into pairs based on school size. At each school, a third and a fourth grade class participated, of which one grade was assigned to the intervention condition and the other grade served as control group. For one of the paired schools, it was randomly determined which intervention (aerobic or cognitively-engaging) would be implemented,

and which grade would serve as intervention group. The same intervention was assigned to the second school in the pair, but the other grade class would receive the intervention. Intervention assignment was only blinded for research assistants, not for teachers and children, as the interventions implicated changes in the physical education schedule. Two schools withdrew from participation after randomization, but before the start of the pre-test. Eleven schools received the aerobic intervention and eleven schools the cognitively engaging intervention. Figure 1 in Appendix A shows the number of classes and children at each stage of the study. In total, 891 children of 22 primary schools participated. Characteristics of participating children are presented in Table 1. Written informed consent was acquired for all children from their legal guardians. The study was approved by the ethical board of the Vrije Universiteit Amsterdam, the Netherlands (approval number #), and registered in the Netherlands Trial Register (number #).

Interventions

Two fourteen-week interventions were designed by researchers (experts in Human Movement Sciences) and Physical Education teachers: an aerobic intervention and cognitively engaging intervention, each consisting of four lessons per week, thereby doubling the number of physical education lessons children were exposed to. The 14 week intervention period was chosen for feasibility reasons (this precisely fitted within a primary school year semester), and because previous studies using similar intervention periods have resulted in positive effects as well (De Greeff et al., 2018). All intervention lessons had a predetermined duration of 30 minutes. The focus of the aerobic intervention was on MVPA, aiming to elicit high heart rate levels to promote children's aerobic fitness via playful forms of aerobic exercise that were highly repetitive and automated, for example, relays, running or individual exercises such as doing squats.The cognitively engaging intervention focused on challenging cognition and motor skills via games (e.g., dodgeball and soccer) and exercises (e.g., balancing, throwing, and catching) that required complex coordination of movements, and that included complex and fast-changing rules to engage children's cognitive skills (Best, 2010). A description of the lesson plan of both interventions is provided in Appendix B (see Table B1 for the aerobic intervention and Table B2 for the cognitively-engaging intervention). The interventions were delivered by external physical education teachers, hired for the project, during regular and extra physical education lessons for 14 weeks in the school year 2016/2017. Schools were free to schedule the lessons in a way that was convenient for them, as long as the lessons were scheduled on four different days. The days and times at which the lessons were provided therefore differ over schools. Teachers were instructed on how to implement the interventions during a training session of 3 h, led by the intervention developers, during which they were familiarized with the goals and the content of the interventions. In addition, they were provided with a manual including a detailed description of each intervention lesson. Observations on intervention implementation were conducted at least two times in each class, after which feedback was provided to the intervention teacher. Control groups followed their regular physical education lessons twice a week.

During two lessons (one in the first week and one in the last week of the intervention), intensity of the lessons was monitored in all three groups. Lessons were chosen based on representativeness of the lesson for the intervention. All children wore an accelerometer (ActiGraph GT3x+, Pensacola, FL, USA) on their right hip to measure the intensity with which they participated. Mean time in MVPA (in minutes) over the two lessons was calculated (see Appendix C). For the intervention groups, volume of MVPA was calculated by multiplying the mean time in MVPA over the two lessons by the number of intervention lessons followed. This was not done for the control group, as no information was available on the number of lessons that children in the control group followed.

Outcomes

Before and after the interventions, all children were tested on academic achievement during the school hours at their own school. The tests that were used are part of a standardized test battery used by most primary schools in the Netherlands, which has been tested on reliability and validity in a large sample of Dutch primary school students (Tomesesen, Weekers, Hilte, Jolink, & Engelen, 2016a; Hop et al., 2016; Tomesen, Wouda et al., 2016). Tests were grade-appropriate, with third grade students making an easier version of the tests than fourth grade students. All tests were conducted following standardized protocols. The reading and mathematics tests were administered by instructed research assistants. The spelling test (a dictation) was conducted by the children's teacher, as children were already familiarized with their teacher's voice and pronunciation. For all tests, the number of correctly answered questions was used as score of academic ability.

Table 1. Baseline characteristics of children, for the total sample and separately for the control group, aerobic intervention group and cognitively engaging intervention group.

	Total sample (n = 891)	Control group (n = 430)	Aerobic intervention group (n = 221)	Cognitively-engaging intervention group (n = 240)
Grade, n grade 3 (%)	456 (51.2)	235 (54.7)[a]	98 (40.0)[a]	123 (56.9)[a]
Gender, n boys (%)	440 (49.4)	219 (51.0)	114 (46.5)	107 (49.5)
Age, in years (SD)	9.17 (0.66)	9.15 (0.67)[b]	9.33 (0.64)[b]	9.06 (0.60)[b]
SES[c] (SD)	4.50 (1.02)	4.49 (1.03)	4.42 (0.93)	4.61 (1.09)

[a]Compared to what could be expected based on chance, there was a higher percentage of third grade children in the control group, and a higher percentage of fourth grade children in the aerobic intervention group.

[b]Controlling for grade level, children in the aerobic intervention were significantly older than children in the control group (p =.02) and the cognitively engaging intervention (p <.001), and children in the cognitively engaging intervention were significantly younger than children in the control group (p <.001).

[c]SES = socioeconomic status; mean parental education level of both parents, measured via a parent questionnaire, ranging from no education (0) to postdoctoral education (7).[24] In case only one of the parents' educational levels was specified, this was used as a measure of the child's SES.

In the reading comprehension, test children read several types of texts (e.g., narrative or informative) and answered 25 multiple choice questions about those texts. Reliability (test–retest reliability = 0.90) and validity of the reading test are good (Tomesen et al., 2016a). The mathematics test consists of 20 questions measuring general mathematics ability. Assignments include basic arithmetic operations and mathematical problems that have to be extracted from text. Reliability (test–retest reliability >0.90) and validity of the mathematics test are good (Hop et al., 2016). The spelling test consists of a dictation in which the teacher reads out a sentence and repeats one word out of that sentence (25 words in total), which children have to correctly write down. Reliability (test–retest reliability >0.90) and validity of the spelling test are good (Tomesen et al., 2016b). The tests were administered in a fixed order, on three different days, within a time frame of 2 weeks.

Statistical analyses

Baseline differences between the three groups were examined using χ2-analyses (grade and gender) or Analysis of Variance (ANOVA; age, SES, and academic achievement) and follow-up analyses with Bonferroni-correction in IBM SPSS Statistics 25.0. In addition, pre-test post-test differences in academic achievement for the three groups were examined using paired t-tests. Subsequently, multilevel path models using maximum likelihood estimation with robust standard errors (MLR) were built in Mplus 7.31 (Muthén & Muthén, 1998–2006) to examine the intervention effects. Intervention effects were examined by relating two dummy variables, contrasting the aerobic intervention group to the control group (1), and the cognitively engaging intervention group to the control group (2), to post-test scores in reading, mathematics, and spelling. The covariates pre-test score, SES, age, gender, and grade were related to academic achievement pre-test and post-test. Covariances were added between scores in reading, mathematics, and spelling, both at pre-test and at post-test. Further, covariances between age and SES, and between age and grade were entered, because of significant relations between these covariates.

For the two intervention groups, volume of MVPA was added as predictor of post-test academic achievement scores. Gender and condition (i.e., aerobic intervention group, or cognitively engaging intervention group) were related to MVPA in this model, as differences between boys and girls, and between intervention groups were expected. An interaction term between group and volume of MVPA was added in a follow-up analysis, to examine whether this relation differed for the two intervention conditions. Lastly, the model was analysed with an interaction between pre-test scores and the dummy variables for condition to examine whether children's initial achievement level was related to the intervention effects. To improve model fit, scores were mean centred, and covariances between the interaction terms and corresponding pre-test scores, and between spelling post-test and reading pre-test were added.

The root mean square error of approximation (RMSEA), comparative fit index (CFI), and standardized root mean square residual (SRMR) were used to evaluate model fit, with cut-offs of 0.06, 0.90, and 0.08, respectively (Hu & Bentler, 1999). Standardized estimates of path coefficients (βs) and corresponding *p*-values were obtained for significance testing. For the models examining the relations with MVPA and baseline performance, only significant relations ($p < .05$) are reported in the manuscript, with further results in Appendix D.

Results

Baseline characteristics

There was an unequal distribution of conditions over grades ($\chi2$ (2) = 6.21, p = .045; see Table 1) and, consequently, there were significant age differences between the groups (F (2, 888) = 10.96, $p < .001$; see Table 1). Children in the three groups did not significantly differ on socioeconomic status (SES) and gender. Table 2 presents mean pre-test and post-test scores for reading, mathematics, and spelling for the three groups. Controlling for grade level, scores in reading, mathematics, and spelling did not significantly differ at pre-test (F(6, 1650) = 1.34, p = .23).

Intervention effects

The model examining the intervention effects on post-test scores in reading, mathematics, and spelling had a good fit to the data (χ^2 (21) = 48.00, RMSEA = 0.04, CFI = 0.99, SRMR = 0.05). The dummy variable contrasting the aerobic intervention group to the control group was not significantly related to post-test reading achievement (β = 0.02 (0.03), p = .60, 95% CI [−0.04 to 0.07]), mathematics achievement (β = −0.02 (0.03), p = .58, 95% CI [−0.08 to 0.04]), or spelling achievement (β = 0.01 (0.03), p = .77, 95% CI [−0.05 to 0.07]). The dummy variable contrasting the cognitively engaging intervention group to the control group was not significantly related to

Table 2. Average pre-test and post-test scores (n correct) and standard deviations in reading, mathematics and spelling for the three conditions.

	n	Control group	*n*	Aerobic intervention group	*n*	Cognitively-engaging intervention group
Reading						
Pre-Test	401	18.22 (4.74)	212	18.82 (4.25)	239	17.85 (5.25)
Post-Test	417	19.44 (4.69)*	214	19.82 (4.30)*	237	19.84 (4.51)*
Mathematics						
Pre-Test	419	14.44 (4.15)	221	14.51 (4.43)	238	14.34 (4.45)
Post-Test	414	15.79 (3.83)*	214	15.59 (3.79)*	237	15.84 (3.75)*
Spelling						
Pre-Test	416	18.31 (5.25)	214	18.47 (4.94)	240	17.77 (5.72)
Post-Test	413	20.30 (4.26)*	215	20.44 (4.26)*	236	19.57 (5.11)*

*Significant difference with pre-test performance, p <.001

post-test reading achievement (β = 0.04 (0.03), p = .19, 95% CI [−0.03 to 0.09]), mathematics achievement (β = −0.004 (0.03), p = .88, 95% CI [−0.08 to 0.04]), or spelling achievement (β = −0.04 (0.03), p = .13, 95% CI [−0.10 to 0.01]).

Moderate-to-vigorous physical activity

As a second aim, volume of MVPA was related to the effectiveness of the interventions. Volume of MVPA significantly differed between the two intervention groups (t (406) = 9.95, p < .001, 95% CI [1.85 to 2.76]), with a significantly higher volume of MVPA in the aerobic intervention group (mean = 9.3 hours, SD = 2.5) than in the cognitively engaging intervention group (mean = 7.0 hours, SD = 2.14). A relation between condition and MVPA was added to control for this difference.

A model with an added relation between volume of MVPA and academic achievement post-test scores resulted in an adequate fit (χ^2 (21) = 55.49, RMSEA = 0.06, CFI = 0.98, SRMR = 0.07). Volume of MVPA was positively related to post-test mathematics achievement (β = 0.09 (0.04), p = .02, 95% CI [0.02 to 0.17]), see Figure 1. This relation did not differ between the two groups (β = .07 (0.13), p = .60, 95% CI [−0.19 to 0.33]).

A significant interaction between volume and group was found for spelling (β = 0.24 (0.10), p = .012, 95% CI [0.05 to 0.43]); volume of MVPA was positively related to post-test spelling achievement in the cognitively engaging intervention group, but not in the aerobic intervention group, see Figure 2.

Baseline achievement

The third aim was to examine whether children's prior level of achievement was related to the intervention effects. The model had an adequate fit to the data (χ^2 (62) = 236.72, RMSEA = 0.06, CFI = 0.95, SRMR = 0.09). Children with lower performance in reading at baseline performed better in reading at the post-test in the cognitively engaging intervention group than in the control group (β = −0.06 (0.03), p = .03, 95% CI [−0.11 to −0.01]), see Figure 3.

Discussion

This study is the first to directly compare the effects of two types of physical activity on academic achievement, one focused on aerobic and one on cognitively engaging physical activity. The interventions did not have significant effects on primary school children's reading, mathematics, or spelling performance. Importantly, there were indications of dose–response effects, as children who were exposed to a higher volume of MVPA performed better in mathematics at the post-test in both intervention groups, and had better post-test spelling achievement in the cognitively engaging intervention group specifically. Further, effects of the cognitively engaging intervention depended on children's initial achievement level, with better post-test reading achievement for lower achievers in reading.

Dose–response effects

Corroborating previous results showing that the effectiveness of physical activity interventions on academic achievement remains yet inconclusive (Donnelly et al., 2016; Singh et al., 2018), we did not find overall effects of the physical activity interventions. A dose–response effect was found, however, which is in line with earlier research (Davis et al., 2011; Mura et al., 2015), suggesting that independent of the *type* of activities involved, a high enough volume of MVPA is necessary to positively affect academic achievement, at least in mathematics. However, although this type of physical activity focusing solely on MVPA had positive effects on mathematics, we found positive effects on both mathematics and spelling for physical activity combining MVPA with cognitive engagement. It thus seems that a combination of MVPA and cognitively

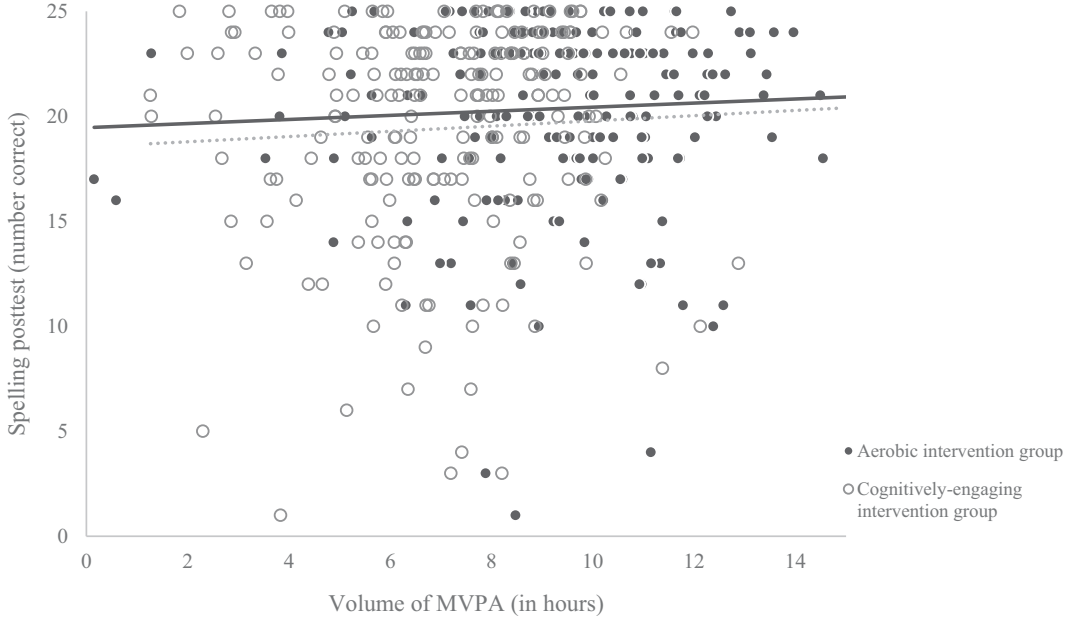

Figure 2. The relation between volume of MVPA and spelling post-test scores for the two intervention groups.

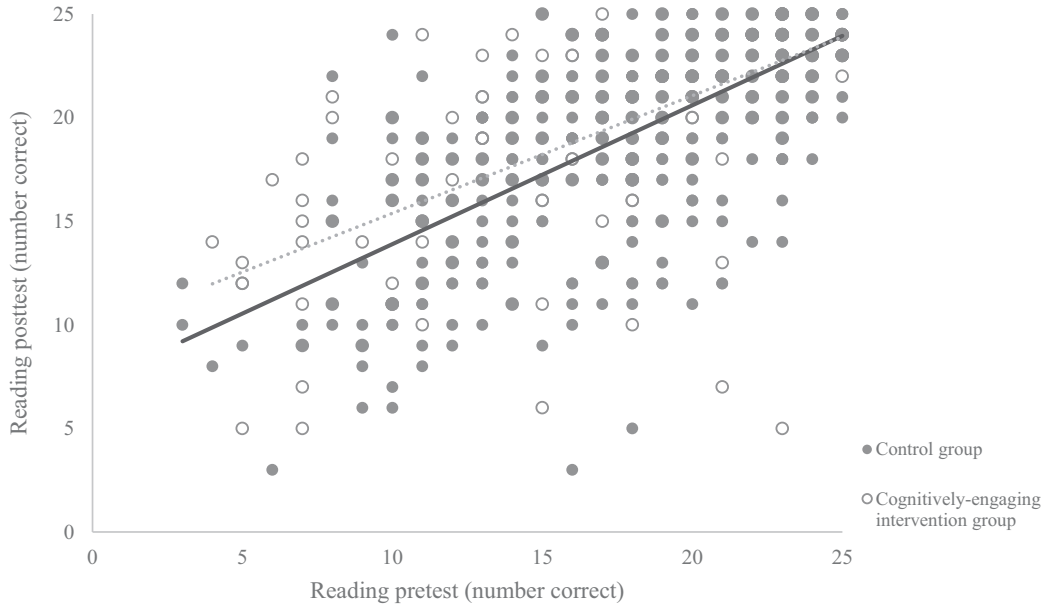

Figure 3. Relation between reading pre-test and reading post-test for the control group and the cognitively engaging intervention group.

engaging physical activity has the most spread-out effects on academic achievement, suggesting that it is important to consider both quantitative and qualitative aspects of physical activity when aiming to improve academic achievement. This same conclusion was reached in a recent study, in which an intervention consisting of team games (targeting cognition and inducing MVPA) had stronger effects on cognition than a regular physical education program and a program focusing on physical exertion (Schmidt et al., 2015).

Following this conclusion, it is likely that there are different mechanisms that can explain the effects of physical activity on academic achievement simultaneously. These might be related to the neurobiological effects of physical activity: changes in brain structure and functioning as a result of aerobic physical activity (Best, 2010; Lubans et al., 2016), and the co-activation of brain areas needed for academic tasks during cognitively engaging physical activity (Álvarez-Bueno et al., 2017; Lubans et al., 2016). Additional mechanisms might be at play at the same time, such as behavioural mechanisms (e.g., improved on task-behaviour, better sleep patterns) or psychosocial mechanisms (e.g., improved self-esteem, increased school engagement) (Bailey, 2017; Lubans et al., 2016). As the intervention programs in the present study focused on either MVPA or cognitive engagement, we are not able to formulate conclusions on the exact mechanisms that can explain the effects of physical activity on academic achievement. To get more insight into this, it would be interesting for future research to further examine whether physical activity that combines MVPA and cognitive engagement has the most beneficial effects on academic achievement. The results presented here strongly point in this direction.

Specificity of intervention effectiveness

The specific results for the different academic domains can possibly be explained by the underlying skills needed to perform well in the specific domains. Spelling performance mainly relies on automatized skills (Farrington-Flint, Stash, & Stiller, 2008). Automatization of skills can be considered an important factor for the intensity with which children participated in the cognitively engaging intervention. That is practising complex skills such as those included in the cognitively engaging intervention is difficult at a high intensity level due to the high cognitive load associated with complex skills (Sweller et al., 2011). With enough training, these skills will become more automated however (Anderson, 1982; Fitts, 1964), reducing cognitive load, and making it possible to practice at a higher intensity. Children who were exposed to a higher volume of MVPA in the cognitively engaging intervention will therefore have automated the complex skills to a larger extent, possibly being beneficial for their spelling performance. Mathematics relies on a combination of complex skills and automatization (Geary, 2004), suggesting that both a high enough intensity and cognitive engagement can result in improved academic performance. For future studies, it seems important to further examine these hypotheses by focusing on how physical activity affects achievement in the different academic domains.

As expected, the effects of the cognitively engaging intervention differed depending on children's baseline academic performance, with better post-test performance for lower-achieving children in reading. Motor skills (the focus of the cognitively engaging intervention in the present study) have already been related to reading comprehension (De Bruijn et al., 2019). This relation was explained by the similarity of both skills, in that they are complex skills requiring controlled and effortful processing. Lower-achieving children are more likely to benefit from this type of intervention, as they have most room for improvement (Diamond & Lee, 2011; Drollette et al., 2014; Resaland et al., 2016; Sibley & Beilock, 2007).

Strengths, limitations, and directions for future research

Strengths of this study include the large sample size, the design, and the use of standardized tests. An important

limitation is that the interventions changed both the volume and the content of physical education lessons. Therefore, no definite conclusions can be drawn about whether it was the type or the volume of physical activity, or a combination, that caused the effects. For future studies, it is important to change one of the parameters at a time in order to disentangle the effects of type compared to volume of physical activity. Still, as the volume of physical activity was enhanced by using two different types of physical activity, it is possible to directly compare the effects of these two interventions. As a second limitation, MVPA was only recorded in two of the 56 lessons. It can therefore be questioned whether this measure adequately reflects the volume of MVPA during the interventions. Yet, as the intervention lessons during which MVPA was measured were chosen based on representativeness of the interventions, it is expected that the measurement of MVPA presents a valid reflection of the volume of MVPA children engaged in during the interventions. Lastly, the amount of cognitive engagement during the interventions was not measured, making it difficult to validate the implementation of the cognitively engaging intervention. Studying children's cognitive engagement is challenging in practice, however (Sinatra et al., 2015), which is why it was chosen not to include this measure in the present study. For future studies, it seems important to find reliable instruments and procedures to tackle this issue.

Conclusion and perspective

This study found no significant effects of two physical activity interventions on academic achievement, a conclusion that corroborates existing literature in which mixed findings on the effectiveness of physical activity are reported (Álvarez-Bueno et al., 2017; Donnelly et al., 2016; Singh et al., 2018). Most importantly, the results support previous conclusions that spending more time on physical activity during the school day does not go at the expense of academic achievement (Donnelly et al., 2016). Even better, it seems that physical activity can have beneficial effects on children's academic achievement, as long as the content of the activities involved is taking into account. Although not explicitly studied, the results presented here suggest that activities that combine a moderate-to-vigorous intensity level with cognitive engagement will have the most beneficial effects on academic achievement. These are important issues for further research, as the effects of physical activity extend way beyond the academic domain, being important for amongst other children's physical fitness, motor skill development, and health and wellbeing (Kohl & Cook, 2013).

Disclosure statement

The authors have no conflicts of interest relevant to this article to disclose.

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Clinical Trial Registry

Learning by moving, registration number NTR5341.

Ethical approval number

VCWE-S-15-00197

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Appendix A: Inclusion flowchart

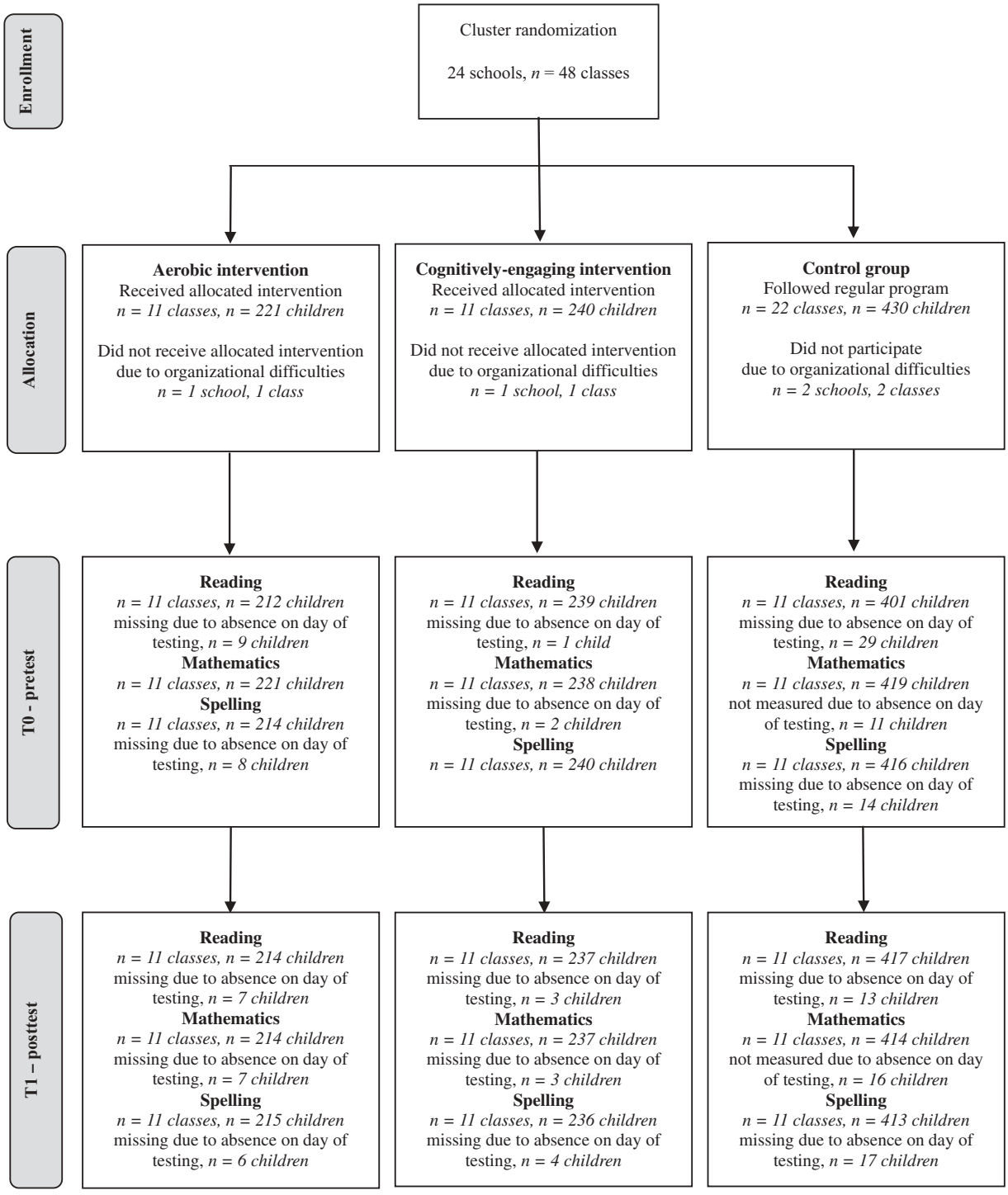

Figure A1. Flow chart with the number of participating classes and children in each stage of the study.

Note: by using Full-Information Maximum Likelihood (FIML) estimation in Mplus, all cases could be included in the analyses.

Appendix B: Session plans describing the intervention lessons for both interventions

Table 1. Session plan for the aerobic intervention program.

	Lesson 1	Lesson 2	Lesson 3	Lesson 4
Week 1	1. Introduction lesson	2. Circuit training	3. Relays: plank, dizzie, build	4. Insanity endurance
Week 2	5. Boxing	6. Rope games	7. HIIT	8. Running
Week 3	9. Bingo relay	10. Insanity endurance	11. Tag game	12. Freerunning
Week 4	13. Shorttrack relay	14. Insanity abs	15. Judo	16. Running relay
Week 5	17. Boxing	18. Bootcamp	19. Robbing game	20. Insanity endurance
Week 6	21. Athletics group camp	22. Pyramid run	23. Dice game	24. Tag games
Week 7	25. Judo	26. Bootcamp	27. Mastermind	28. HIIT
Week 8	29. Circuit in pairs	30. Boxing	31. Running games relay	32. Athletics group camp
Week 9	33. Insanity endurance	34. Freerunning	35. Bingo relay	36. Rope lesson
Week 10	37. Circuit in pairs	38. Insanity endurance	39. Judo	40. HIIT
Week 11	41. Circuit training	42. Boxing	43. Insanity abs	44. Tag games
Week 12	45. Circuit training	46. *	47. *	48. *
Week 13	49. *	50 *	51. *	52. *
Week 14	53. *	54. *	55. *	56. *

Note. Only 45 lessons were planned, the other 11 lessons were open for catching up lessons that were cancelled, or for repeating lessons.

Table 2. Session plan for the cognitively engaging intervention program.

	Lesson 1	Lesson 2	Lesson 3	Lesson 4
Week 1	1. Introduction lesson	2. Circuit training	3. Relays: plank, dizzie, build	4. Insanity endurance
EFs		*WM, inhibition*	*WM, strategic play*	*Inhibition*
Motor skills		*Bilateral coordination (running)*	*Bilateral coordination (moving sideways, balance)*	*Ball skills (throwing, catching, shooting, dribbling)*
Week 2	5. Ball game	6. Relay game	7. Hoepla	8. Bootcamp
EFs	*WM, inhibition*	*WM, inhibition*	*WM*	*WM*
Motor skills	*Ball skills (throwing, catching)*	*Bilateral coordination (turning)*	*Ball skills: throwing, catching, aiming*	*Bilateral coordination*
Week 3	9. Dodgeball	10. Basketball	11. Ball relay game	12. Baseball
EFs	*Inhibition, strategic play*	*Inhibition*	*Strategic play*	*WM*
Motor skills	*Ball skills (throwing, catching)*	*Ball skills (throwing, catching, shooting, dribbling)*	*Ball skills (throwing, catching, rolling), balance*	*Ball skills (throwing, catching), bilateral coordination*
Week 4	13. Moving to music	14. Chaotic matt game	15. James Bond	16. Team work
EFs	*Inhibition*	*Strategic play*	*Strategic play*	*Strategic play*
Motor skills	*Balance, bilateral coordination, rhythm*	*Ball skills (throwing, catching, rolling), balance*	*Ball skills (throwing, catching), bilateral coordination (dodging)*	*Bilateral coordination, balance*
Week 5	17. Pinball	18. Swedish running game	19. Kinging	20. Climbing and clambering
EFs	*Inhibition, shifting*	*WM*	*Inhibition*	*Obeying rules*
Motor skills	*Ball skills (rolling, throwing), bilateral coordination (running)*	*Bilateral coordination (running)*	*Ball skills (throwing, catching)*	*Bilateral coordination (climbing), balance*
Week 6	21. Football	22. Robbing quartet	23. Circulation volleyball	24. Balancing
EFs	*WM, inhibition*	*WM, inhibition*	*WM, inhibition*	*Obeying rules*
Motor skills	*Ball skills (passing, dribbling, shooting)*	*Coordination*	*Ball skills (throwing, catching)*	*Balance, ball skills (football), coordination*
Week 7	25. Juggling	26. Search for the king 2	27. Expedition Robinson 2	28. Ball game 2
EFs	*Inhibition*	*WM, inhibition*	*WM*	*WM, inhibition*
Motor skills	*Hand-eye coordination*	*Coordination (running), balance*	*Balance, hand-eye coordination*	*Ball skills*
Week 8	29. Relay game 2	30. Olympia 2	31. Hoepla 2	32. Bootcamp 2
EFs	*WM, inhibition*	*WM, inhibition*	*WM*	*WM, inhibition*
Motor skills	*Bilateral coordination (turning)*	*Ball skills*	*Bilateral coordination, eye-hand coordination*	*Bilateral coordination, balance*
Week 9	33. Dodgeball 2	34. Basketball 2	35. Ball relay games 2	36. Baseball 2
EFs	*Cooperation, strategic play*	*WM*	*WM, inhibition*	*WM, inhibition*
Motor skills	*Ball skills (throwing, catching)*	*Ball skills (catching, throwing, dribbling)*	*Balance, bilateral coordination (moving sideways)*	*Ball skills*
Week 10	37. Moving to music 2	38. Chaotic matt game 2	39. James Bond 2	40. Team work 2
EFs	*WM, inhibition*	*Inhibition*	*Strategic play*	*Strategic play*
Motor skills	*Balance, bilateral coordination, rhythm*	*Ball skills*	*Ball skills (throwing, catching), bilateral coordination (dodging)*	*Bilateral coordination, balance*
Week 11	41. Pinball 2	42. Swedish running game	43. Kinging	44. Climbing and clambering 2
EFs	*Inhibition, shifting*	*WM*	*Inhibition*	*Obeying rules*
Motor skills	*Ball skills (rolling, throwing), bilateral coordination (running)*	*Bilateral coordination (running)*	*Ball skills (throwing, catching)*	*Bilateral coordination (climbing), balance*
Week 12	45. Football 2	46. Robbing quartet 2	47. Circulation volleyball	48. Balancing 2
EFs	*WM*	*WM, inhibition*	*WM, inhibition*	*Obeying rules*

(*Continued*)

Table 2. (Continued).

	Lesson 1	Lesson 2	Lesson 3	Lesson 4
Motor skills	*Ball skills*	*Coordination*	*Ball skills (throwing, catching)*	*Balance, ball skills (football), coordination*
Week 13	49. Juggling 2	50. Field ball game	51. Potpourri 1	52. Potpourri 2
EFs	*Inhibition*	*Inhibition, strategic play*	*Inhibition*	*WM*
Motor skills	*Ball skills, hand-eye coordination*	*Ball skills (throwing, catching), bilateral coordination*	*Ball skills (throwing, catching, defending), bilateral coordination*	*Ball skills (throwing, catching, dribbling), bilateral coordination*
Week 14	53. *	54. *	55. *	56. *

Note. EFs = executive functions; WM = working memory; * The last four lessons were open for lessons that were cancelled, or for repeating lessons.

Appendix C: Calculation of MVPA

In all three groups, MVPA was measured during two physical education lessons using accelerometers (ActiGraph GT3x+, Pensacola, FL, USA). The accelerometer was attached to the child's right hip using an elastic belt. Accelerations in three directions were measured with a frequency of 100 Hz. Data analyses were done in the software ActiLife (v6.8.2). Only data of the vertical axis were used for analysis. An epoch length of 1 second was chosen (Trost et al., 2011). The cut-off points used to determine the number of counts per minute were as follows: moderate: 2296–4011 counts/min; vigorous: >4012 counts/min (Evenson et al., 2008). As a measure of MVPA, time spent at a moderate and at a vigorous intensity level (in minutes) was summed and averaged over the two lessons.

Table 100.1. presents the average intensity per group. Intensity of the physical education lessons differed between the three groups (F (2, 806) = 45.81, p <.001), with a higher intensity in the aerobic intervention (M = 12.36, sd = 3.08) than in the cognitively engaging intervention (M = 9.29, sd = 2.47, p <.001) and the control condition (M = 10.65, sd = 3.70, p <.001). The intensity of the cognitively engaging intervention was lower than that of the control group (p <.001).

Table 100. Average amount of MVPA in the three groups.

	Mean MVPA (in minutes) (*sd*)	*n*	Volume (dose * number of lessons of MVPA (*sd*)	*n*
Control group	10.65 (3.7)	395	-	-
Aerobic intervention group	12.36 (3.08)	205	558.69 (151.8)	202
Cognitively-engaging intervention group	9.29 (2.5)	207	420.30 (128.6)	206

Appendix D: Analysis on MVPA and baseline academic achievement

D.1. Results of the models examining the relation between MVPA and intervention effects on academic achievement

A model with an added relation between volume of MVPA and academic achievement post-test scores resulted in an adequate fit (χ (21) = 55.49, RMSEA = 0.06, CFI = 0.98, SRMR = 0.07). Volume of MVPA was positively related to post-test mathematics achievement (β = 0.09 (0.04), p =.02, 95% CI [0.02 to 0.17]). This relation was not found for reading (β = −0.04 (0.05), p =.51, 95% CI [−0.14 to 0.07]) nor for spelling (β = 0.002 (0.04), p =.96, 95% CI [−0.07 to 0.07]).

In a follow-up analysis, an interaction term between volume of MVPA and condition was added. This model proved to have an adequate fit to the data (χ (27) = 88.78, RMSEA = 0.08, CFI = 0.96, SRMR = 0.06). Although the RMSEA was above the predetermined cut-off value, we still decided to use the model, as all other values were acceptable. The interaction between volume of MVPA and condition was significantly related to post-test achievement in spelling (β = 0.24 (0.10), p =.01, 95% CI [0.05 to 0.43]). No relation was found with post-test achievement in reading (β = 0.05 (0.17), p =.78, 95% CI [−0.29 to 0.38]) or mathematics (β = 0.07 (0.13), p =.60, 95% CI [−0.19 to 0.33]), indicating that volume of MVPA was not differently related to post-test achievement in reading or mathematics for the two interventions.

D.2. Results of the model examining interactions between children's initial level of achievement and intervention

The third aim of this study was to examine whether children's prior level of achievement was related to the intervention effects. The model with an added interaction between pre-test scores and the dummy variables for condition had an adequate fit to the data (χ (62) = 236.72, RMSEA = 0.06, CFI = 0.95, SRMR = 0.09).

Children with lower performance in reading at baseline performed better in reading at the post-test in the cognitively engaging intervention group than in the control group (β = −0.06 (0.03), p =.03, 95% CI [−0.11 to −0.01]), see Figure 4. No significant relation was found for the interaction between the dummy variable contrasting the cognitively engaging intervention group and the control group and baseline mathematics performance (β = −.03 (.04), p =.37, 95% CI [−.11 to.04]), or baseline spelling performance (β =.07 (.04), p =.06, 95% CI [−.01 to.14]).

The interaction term between baseline performance and the dummy variable contrasting the aerobic intervention and the control group was not significant for reading (β = −.01 (.03), p =.75, 95% CI −.07 to.05]), mathematics (β = −.01 (.04), p =.82, 95% CI [−.09 to.07]), or spelling (β =.03 (.03), p =.37, 95% CI [−.03 to.10]), indicating that the effectiveness of the aerobic intervention did not differ depending on baseline academic performance.